PHB221 · Fall 2026

Molecular Genetics

Question bank — 140 multiple-choice questions across 7 lectures. Answer key with explanations at the end.

Lecture 1 — DNA structure and replication

1. How many hydrogen bonds hold a G–C base pair together?
A. 1
B. 2
C. 3
D. 4

2. Which enzyme removes the RNA primers during DNA replication?
A. DNA polymerase III
B. Ligase
C. Primase
D. DNA polymerase I

3. Okazaki fragments are formed on the:
A. Lagging strand
B. Leading strand
C. Template strand of RNA
D. Coding strand

4. A nucleoside consists of:
A. Base only
B. Base + pentose
C. Base + pentose + phosphate
D. Pentose + phosphate

5. Which DNA form is left-handed?
A. A-DNA
B. Z-DNA
C. B-DNA
D. All of them

6. Which protein relieves the supercoils that form ahead of the replication fork?
A. SSB proteins
B. DnaA
C. Topoisomerase
D. Primase

7. Rosalind Franklin's X-ray work showed that the sugar–phosphate backbone is:
A. Inside the helix
B. Only on one strand
C. Absent in DNA
D. Outside the helix

8. A DNA sample contains 30% guanine. What percentage is thymine?
A. 30%
B. 20%
C. 40%
D. 70%

9. Chargaff's rule states that:
A. A = G and C = T
B. Purines = pyrimidines
C. A + T = G + C always
D. DNA has no uracil

10. The bond joining a nitrogenous base to the pentose sugar is the:
A. N-glycosidic bond
B. Hydrogen bond
C. Phosphodiester bond
D. Peptide bond

11. A phosphodiester bond links:
A. The 3'-OH of one sugar to the 5' carbon of the next via phosphate
B. Two bases across the helix
C. Base to sugar
D. Two phosphates directly

12. Which DNA form exists under dehydrating conditions?
A. B-DNA
B. Z-DNA
C. A-DNA
D. C-DNA

13. About how many base pairs wrap around one histone octamer in a nucleosome?
A. 46
B. 92
C. 146
D. 1,000

14. Why are replication origins AT-rich?
A. A=T pairs have only 2 hydrogen bonds, so they melt easily
B. AT is needed by ligase
C. Primase only reads AT
D. GC is absent in eukaryotes

15. Eukaryotes differ from prokaryotes in replication because they have:
A. Multiple origins
B. A single origin
C. No primers
D. No lagging strand

16. Which protein binds the origin first and needs ATP to start melting it?
A. Pol I
B. SSB
C. Ligase
D. DnaA

17. Single-stranded DNA binding (SSB) proteins:
A. Synthesize primers
B. Relieve supercoils
C. Seal nicks
D. Keep separated strands apart and protect them from nucleases

18. The RNA primer made by primase is about how long?
A. 2 nucleotides
B. 10 nucleotides
C. 100 nucleotides
D. 1,000 nucleotides

19. DNA polymerase III reads the template in which direction?
A. 5'→3'
B. It doesn't read a template
C. Both directions equally
D. 3'→5'

20. The leading strand needs how many primers?
A. One
B. None
C. One per Okazaki fragment
D. Two

21. Proofreading by DNA polymerase III uses its:
A. 5'→3' polymerase
B. 5'→3' exonuclease
C. 3'→5' exonuclease
D. Ligase activity

22. The energy for DNA ligase to seal a nick comes from:
A. GTP → GDP
B. ATP → AMP + PPi
C. NADH
D. Light

23. Semiconservative replication means each daughter DNA has:
A. Two new strands
B. One old and one new strand
C. Two old strands
D. Mixed fragments on both strands

Lecture 2 — End replication problem and transcription

24. Telomerase extends telomeres because it acts as a:
A. DNA-dependent RNA polymerase
B. Reverse transcriptase with its own RNA template
C. Ligase
D. Restriction enzyme

25. RNA polymerase III transcribes:
A. All mRNA
B. rRNA except 5S
C. Mitochondrial DNA only
D. tRNA and 5S rRNA

26. The coding (sense) strand of DNA:
A. Is used as the template
B. Is complementary to the mRNA
C. Has the same sequence as the mRNA (T for U)
D. Is always on the 3' side

27. Which general transcription factor has helicase activity and phosphorylates the CTD of RNA pol II?
A. TFIIB
B. TFIID
C. TFIIE
D. TFIIH

28. The end replication problem occurs at the:
A. 5' end of the leading strand
B. Origin of replication
C. 3' end of the lagging-strand template of linear chromosomes
D. Middle of the chromosome

29. Why don't bacteria have the end replication problem?
A. They lack DNA polymerase
B. Their chromosome is circular
C. They have no primers
D. They use RNA genomes

30. The human telomere repeat sequence is:
A. TATAAA
B. GCGCGC
C. AAUAAA
D. TTAGGG

31. How many telomeres does a normal human somatic cell have?
A. 23
B. 46
C. 92
D. 184

32. Telomerase is best described as:
A. A reverse transcriptase carrying its own RNA template
B. A DNA-dependent RNA polymerase
C. A restriction enzyme
D. A helicase

33. The correct order of telomerase action is:
A. Translocation → binding → polymerization
B. Binding → translocation → ligation
C. Polymerization → binding → translocation
D. Binding → polymerization → translocation

34. Telomere length falls from about ___ at birth to about ___ late in life.
A. 1,000 to 500 bp
B. 10,000 to 5,000 bp
C. 100,000 to 50,000 bp
D. 5,000 to 1,000 bp

35. Cancer cells typically show:
A. Low telomerase and short telomeres
B. No cell division
C. No telomeres
D. High telomerase and maintained telomeres

36. The 2009 Nobel Prize (telomeres and telomerase) went to:
A. Blackburn, Greider and Szostak
B. Watson and Crick
C. Ambros and Ruvkun
D. Kaelin, Ratcliffe and Semenza

37. Template strand 3'-TACGGA-5' gives which mRNA?
A. 5'-ATGCCT-3'
B. 5'-AUGCCU-3'
C. 5'-UACGGA-3'
D. 3'-AUGCCU-5'

38. Which RNA polymerase makes tRNA?
A. Pol I
B. Pol II
C. Pol III
D. Primase

39. Intron sequences start with ___ and end with ___.
A. AG … GT
B. TATA … AAUAAA
C. GT … AG
D. AUG … UGA

40. The CAAT box mainly controls:
A. Where translation starts
B. Splicing
C. Polyadenylation
D. How often transcription starts

41. Repressors bind to:
A. Silencers
B. Enhancers
C. TATA box only
D. Poly-A signal

42. A TATA box is an example of a:
A. Trans-acting element
B. Ribozyme
C. Coactivator
D. Cis-acting element

43. Which factor recognizes the core promoter via TBP?
A. TFIID
B. TFIIB
C. TFIIE
D. TFIIF

44. TFIIB's role in initiation is to:
A. Act as a bridge between TFIID and RNA pol II
B. Unwind DNA
C. Add the poly-A tail
D. Phosphorylate histones

45. The poly-A signal in pre-mRNA is:
A. TTAGGG
B. AUG
C. AAUAAA
D. GU…AG

Lecture 3 — RNA processing and translation

46. The 5' cap of eukaryotic mRNA is:
A. 7-methylguanosine
B. A poly-A sequence
C. A TATA box
D. An anticodon loop

47. Which ribosomal site holds the tRNA carrying the growing polypeptide?
A. A site
B. P site
C. E site
D. Exit tunnel

48. Wobble pairing happens at which position of the codon?
A. First base
B. Second base
C. Any base
D. Third base

49. Histone acetyltransferase (HAT) activates genes by:
A. Adding methyl groups to DNA
B. Adding the 5' cap
C. Cutting introns
D. Lowering histone basicity so DNA loosens

50. Which enzyme reverses HAT's effect and silences genes?
A. DNA ligase
B. Primase
C. Histone deacetylase (HDAC)
D. Telomerase

51. The unprocessed eukaryotic transcript is called:
A. cDNA
B. snRNA
C. tRNA
D. hnRNA (pre-mRNA)

52. snRNPs ('snurps') are made of:
A. DNA + histones
B. rRNA only
C. Small nuclear RNA + proteins
D. Lipids

53. Which is NOT a function of the spliceosome?
A. Recognizing intron–exon boundaries
B. Holding pre-mRNA in shape
C. Catalysing intron removal
D. Adding the poly-A tail

54. Alternative splicing allows:
A. One gene to make several proteins
B. DNA replication
C. Translation without ribosomes
D. Removal of exons only

55. The 5' cap is added when the transcript is about:
A. 2–5 nt long
B. 20–25 nt long
C. 200 nt long
D. After transcription is finished

56. Which is a function of the 5' cap?
A. Encodes methionine
B. Protects mRNA from nucleases and helps ribosome binding
C. Signals termination
D. Removes introns

57. The poly-A tail is:
A. Encoded in the gene
B. Added enzymatically after transcription
C. Added by the ribosome
D. Part of tRNA

58. tRNA is roughly how long?
A. 8 nt
B. 80 nt
C. 800 nt
D. 8,000 nt

59. Aminoacyl-tRNA synthetase attaches the amino acid using energy from:
A. ATP
B. GTP
C. NADPH
D. Light

60. At the wobble position, G can pair with:
A. A or G
B. T
C. Only C
D. C or U

61. Ribosomes are composed of:
A. 60% rRNA, 40% protein
B. 100% protein
C. 40% rRNA, 60% protein
D. DNA and protein

62. The E site of the ribosome holds:
A. The incoming aminoacyl-tRNA
B. The empty tRNA about to leave
C. The tRNA with the growing chain
D. The mRNA cap

63. The polypeptide chain grows from:
A. C-terminus to N-terminus
B. The middle outward
C. Both ends
D. N-terminus to C-terminus

64. GTP is used during which steps of elongation?
A. Peptide bond formation only
B. Capping
C. Termination only
D. Codon recognition and translocation

65. At termination, the peptide is released by:
A. A tRNA binding the stop codon
B. Hydrolysis of the bond to the P-site tRNA
C. Ligase
D. Splicing

66. Many ribosomes translating one mRNA together form a:
A. Spliceosome
B. Nucleosome
C. Polysome
D. Proteasome

Lecture 4 — Post-translational modifications and the genetic code

67. At normal oxygen levels, HIF-1α is degraded because it is first:
A. Phosphorylated on tyrosine
B. Acetylated by HAT
C. Prolyl-hydroxylated, then recognized by VHL and ubiquitinated
D. Glycosylated in the Golgi

68. Protein phosphorylation occurs on which residues?
A. Lys and Arg
B. Pro and Lys
C. Ser, Thr and Tyr
D. Gly and Ala

69. How many of the 64 codons specify amino acids?
A. 20
B. 60
C. 61
D. 64

70. The proteome (>1 million proteins) is much larger than the genome (20–25,000 genes) mainly because of:
A. Mutation
B. Telomerase
C. DNA replication errors
D. Alternative splicing plus PTMs

71. Ubiquitination is classified as a PTM by addition of:
A. Chemical groups
B. Complex groups
C. Polypeptides
D. Cleavage

72. Glycosylation belongs to which PTM group?
A. Chemical groups
B. Polypeptides
C. Complex groups
D. Proteolysis

73. Phosphorylation makes a protein region:
A. Positively charged and hydrophobic
B. Uncharged
C. Negatively charged and hydrophilic
D. Lipid-soluble

74. Acetyl-CoA carboxylase is ___ by phosphorylation.
A. Inhibited
B. Activated
C. Degraded
D. Not affected

75. p53 is ___ by phosphorylation of its N-terminus.
A. Activated
B. Inhibited
C. Degraded
D. Exported

76. Hydroxylation of proline and lysine is classically seen in:
A. Haemoglobin
B. Collagen
C. Insulin
D. Histone H1

77. Lysine can be methylated up to:
A. Once
B. Twice
C. Three times
D. Five times

78. Ubiquitin's molecular weight is about:
A. 0.86 kDa
B. 8.6 kDa
C. 86 kDa
D. 860 kDa

79. In a polyubiquitin chain, each new ubiquitin links to:
A. One of seven lysines or the N-terminal Met of the previous ubiquitin
B. Any serine
C. The C-terminus of the target only
D. DNA

80. Removal of the N-terminal methionine is an example of:
A. Phosphorylation
B. Proteolysis
C. Carboxylation
D. Acetylation

81. Gamma-carboxylation of clotting factors needs:
A. Vitamin C
B. Vitamin D
C. Vitamin B12
D. Vitamin K

82. Under low oxygen, HIF-1α:
A. Is hydroxylated and degraded
B. Binds VHL more strongly
C. Accumulates and switches on genes like EPO
D. Is exported from cells

83. Which protein recognizes hydroxylated HIF-1α?
A. p53
B. Dicer
C. TBP
D. VHL

84. Translate: AUG-UUU-GGC-UAA
A. Met-Leu-Gly-Stop
B. Met-Phe-Gly-Stop
C. Met-Phe-Ala
D. Stop-Phe-Gly-Met

85. Which codon is NOT a stop codon?
A. UGG
B. UAG
C. UGA
D. UAA

86. CGA and AGG both code for arginine. This illustrates:
A. Wobble only
B. Redundancy (degeneracy) of the code
C. A mutation
D. Overlapping code

Lecture 5 — Regulation of gene expression I

87. A microRNA silences its target by binding to the:
A. Promoter
B. 5' cap
C. TATA box
D. 3'UTR of the mRNA

88. Methylation of CpG islands near a promoter usually:
A. Activates transcription
B. Represses transcription
C. Has no effect
D. Causes alternative splicing

89. Housekeeping genes encoding metabolic enzymes show:
A. Constitutive expression
B. Facultative expression
C. No expression
D. Only embryonic expression

90. Epigenetics refers to changes in gene expression that:
A. Do not alter the DNA sequence
B. Alter the DNA sequence
C. Only affect RNA viruses
D. Only happen in bacteria

91. C/EBPs are a family of how many transcription factors?
A. Two
B. Four
C. Six
D. Twelve

92. Sp1 binds which promoter element?
A. GC box
B. TATA box
C. CAAT box only
D. Poly-A signal

93. An activator's two domains are:
A. Kinase and phosphatase
B. DNA-binding and regulatory
C. Helicase and primase
D. Cap and tail

94. Coactivators:
A. Transmit signals from activators to basal factors
B. Bind silencers
C. Cut miRNA hairpins
D. Methylate DNA

95. Enhancers far from a gene can act on it because:
A. They encode the gene
B. They are transcribed into mRNA
C. The DNA loops, bringing the enhancer-bound activator near the promoter
D. They move along the chromosome

96. Methylation of CpG islands near a promoter usually:
A. Activates transcription
B. Represses transcription
C. Has no effect
D. Causes splicing

97. Methyl-CpG binding protein silences genes mainly by recruiting:
A. HAT
B. Telomerase
C. RNA pol II
D. HDAC

98. About what fraction of the primary transcript is removed during processing?
A. 10%
B. 50%
C. 90%
D. None

99. miRNA is about how long?
A. 2 nt
B. 22 nt
C. 220 nt
D. 2,200 nt

100. Which enzyme cuts the miRNA hairpin?
A. Dicer
B. Ligase
C. Primase
D. Telomerase

101. miRNA works inside which complex?
A. Spliceosome
B. RISC
C. Proteasome
D. Nucleosome

102. Which polymerase transcribes miRNA genes?
A. RNA pol I
B. RNA pol III
C. RNA pol II
D. DNA pol III

103. Over-expression of miRNA-181 would:
A. Increase telomerase
B. Decrease telomerase
C. Not affect telomerase
D. Lengthen telomeres

104. The 2024 Nobel Prize in Physiology or Medicine was for:
A. microRNA
B. Telomerase
C. HIF-1α
D. CRISPR

Lecture 6 — Regulation of gene expression II

105. Bromodomains recognize:
A. Methylated histone tails
B. Phosphorylated DNA
C. Acetylated histone tails
D. CpG islands

106. Circular RNAs commonly regulate gene expression by:
A. Acting as miRNA sponges
B. Coding for proteins
C. Adding the 5' cap
D. Replacing histone H3

107. Which histone variant replaces H3 at centromeres?
A. H1
B. H2A.Z
C. H4
D. CENP-A

108. circRNAs are formed by:
A. Canonical splicing
B. Back-splicing joining 5' to 3' end
C. Reverse transcription
D. Replication

109. circRNA half-life is about:
A. 4 minutes
B. 48 hours
C. 48 days
D. 4 years

110. Why are circRNAs so stable?
A. They are double-stranded
B. They are in the nucleolus
C. They are methylated
D. They have no free ends for exonucleases

111. lncRNAs are typically:
A. Shorter than 22 nt
B. Circular
C. Longer than 200 nt, often 1,000–10,000 nt
D. Translated into protein

112. An lncRNA that binds a transcription factor and keeps it from its target acts as a:
A. Guide
B. Decoy
C. Scaffold
D. Ribozyme

113. HOTAIR over-expression lowers ubiquitin protein because HOTAIR:
A. Degrades ubiquitin mRNA
B. Is translated into a repressor
C. Recruits HDAC to the ubiquitin promoter
D. Methylates ubiquitin

114. If HOTAIR (which sponges miR-236) decreases, p21 will:
A. Increase
B. Decrease
C. Stay the same
D. Be mutated

115. The linker histone that forms the 30 nm fibre is:
A. H2A
B. H1
C. H3
D. CENP-A

116. Which nucleosome sites are most frequently accessible?
A. Those at the centre
B. None
C. Those near the ends
D. Only histone sites

117. Nucleosome remodeling complexes use energy from:
A. GTP
B. ATP
C. NADH
D. Light

118. Moving a histone octamer to another DNA molecule is called:
A. Sliding
B. Remodeling
C. Transfer
D. Positioning

119. A:T-rich DNA is favoured where the minor groove:
A. Faces the octamer
B. Faces away from the octamer
C. Is absent
D. Is methylated

120. Serine phosphorylation of histone tails gives:
A. Repressed chromatin
B. Active chromatin
C. Heterochromatin always
D. No change

121. Chromodomains recognize:
A. Acetylated tails
B. Methylated tails
C. Phosphorylated DNA
D. CpG islands

122. Which is NOT a level of eukaryotic gene regulation?
A. DNA replication fidelity
B. RNA processing
C. Translation
D. Transcription

123. Which haemoglobin has the highest oxygen affinity?
A. Embryonic
B. Fetal
C. Adult
D. All equal

124. Adult haemoglobin consists of:
A. 2α + 2γ
B. 2α + 2β
C. 2ζ + 2ε
D. 4β

125. Regulatory protein-binding sites near promoters are usually:
A. 1–2 nt
B. 10–15 nt
C. 100–150 nt
D. 1,000 nt

126. DNA-binding proteins usually read the major groove because:
A. It is narrower
B. It contains no bases
C. Individual base pairs can be distinguished there
D. It is single-stranded

127. The leucine zipper is formed by:
A. One α-helix
B. Zinc atoms
C. Two subunits joined by hydrophobic leucines into a Y shape
D. RNA

Lecture 7 — Genetic variation and the Human Genome Project

128. When was the first truly complete (telomere-to-telomere) human genome announced?
A. 2000
B. 2003
C. 2013
D. 2022

129. Which non-human organism was NOT among those sequenced by the HGP?
A. E. coli
B. Baker's yeast
C. Zebrafish
D. Fruit fly

130. How many countries took part in the International Human Genome Sequencing Consortium?
A. 2
B. 6
C. 20
D. 50

131. Which sequencing method did the HGP use (after improving it)?
A. Sanger sequencing
B. Nanopore
C. Illumina
D. Maxam–Gilbert

132. The June 2000 draft contained about how many gaps?
A. Fewer than 400
B. About 4,000
C. None
D. More than 150,000

133. The 2003 'essentially complete' sequence covered about:
A. 50%
B. 90%
C. 92%
D. 100%

134. The hardest regions to sequence were:
A. Exons
B. Promoters
C. Telomeres and centromeres
D. Mitochondrial genes

135. When was the first truly complete human genome announced?
A. April 2003
B. June 2000
C. March 2021
D. 31 March 2022

136. The Egyptian Genome Project was launched in:
A. March 2021
B. 2003
C. 1990
D. 2022

137. Which is an aim of the Egyptian Genome Project?
A. Sequencing the mouse genome
B. Cloning humans
C. Replacing Sanger sequencing
D. Studying genomes of ancient Egyptians

138. Humans are about what percent identical at the base-pair level?
A. 90%
B. 99%
C. 99.9%
D. 100%

139. A single-base variant found in 5% of a population is a:
A. Mutation
B. SNP
C. Deletion
D. Translocation

140. '45 C>T' means:

A. 45 copies of C
B. C and T are deleted
C. At nucleotide 45, normal C is replaced by T
D. A 45-base insertion

Answer key

Lecture 1

1	**C**	**3** — G≡C has 3 hydrogen bonds, which makes it about 50% stronger than A=T (2 bonds).
2	**D**	**DNA polymerase I** — Pol I uses its 5'→3' exonuclease activity to remove primers, then fills the gaps.
3	**A**	**Lagging strand** — The lagging strand grows away from the fork, so it is made discontinuously in short fragments.
4	**B**	**Base + pentose** — Nucleoside = base + sugar. Add a phosphate and it becomes a nucleotide.
5	**B**	**Z-DNA** — Z-DNA is the rare left-handed form. A and B are right-handed.
6	**C**	**Topoisomerase** — Topoisomerases relieve supercoiling caused by unwinding.
7	**D**	**Outside the helix** — The charged backbone faces water on the outside; the hydrophobic bases are inside.
8	**B**	**20%** — G = C = 30%, so G + C = 60%. A + T = 40%, split equally: T = 20%.
9	**B**	**Purines = pyrimidines** — A = T and G = C, so A + G (purines) = T + C (pyrimidines).
10	**A**	**N-glycosidic bond** — Base–sugar = N-glycosidic. Nucleotide–nucleotide = phosphodiester.
11	**A**	**The 3'-OH of one sugar to the 5' carbon of the next via phosphate** — That 3'→5' linkage gives every strand a 5' end and a 3' end.
12	**C**	**A-DNA** — A-DNA is right-handed and dehydrated; B is the common cellular form; Z is left-handed.
13	**C**	**146** — About 146 bp around eight histones form one nucleosome.
14	**A**	**A=T pairs have only 2 hydrogen bonds, so they melt easily** — Fewer hydrogen bonds make AT regions easier to unwind.
15	**A**	**Multiple origins** — Large eukaryotic genomes need many origins to finish replication in time.
16	**D**	**DnaA** — DnaA binds the AT-rich origin sequences and requires ATP.
17	**D**	**Keep separated strands apart and protect them from nucleases** — SSBs coat single strands so they don't re-anneal or get degraded.
18	**B**	**10 nucleotides** — Primase lays down a short primer of about 10 nucleotides.
19	**D**	**3'→5'** — It reads 3'→5' and builds the new strand 5'→3'.
20	**A**	**One** — It is synthesized continuously from a single primer.
21	**C**	**3'→5' exonuclease** — It removes a wrongly added nucleotide from the growing 3' end.
22	**B**	**ATP → AMP + PPi** — Ligase splits ATP to AMP + PPi to form the phosphodiester bond.
23	**B**	**One old and one new strand** — One parental strand is conserved in each new double helix.

Lecture 2

24	**B**	**Reverse transcriptase with its own RNA template** — Telomerase carries an RNA template and synthesizes DNA from it, which is reverse transcription.
25	**D**	**tRNA and 5S rRNA** — Pol I: rRNA (except 5S). Pol II: mRNA. Pol III: tRNA + 5S rRNA.
26	**C**	**Has the same sequence as the mRNA (T for U)** — The template (antisense) strand is read; the coding strand matches the mRNA sequence.
27	**D**	**TFIIH** — TFIIH opens the DNA (helicase) and phosphorylates the CTD, releasing pol II into elongation.
28	**C**	**3' end of the lagging-strand template of linear chromosomes** — After the last primer is removed there's no 3'-OH to fill the gap at the tip.
29	**B**	**Their chromosome is circular** — A circular chromosome has no ends.
30	**D**	**TTAGGG** — Telomeres are tandem TTAGGG repeats with a 12–16 nt 3' overhang.
31	**C**	**92** — 46 chromosomes × 2 ends = 92 telomeres.
32	**A**	**A reverse transcriptase carrying its own RNA template** — It makes DNA from its built-in RNA template.
33	**D**	**Binding → polymerization → translocation** — It binds the overhang, copies its template, then slides along and repeats.
34	**B**	**10,000 to 5,000 bp** — About 10,000 bp at birth, about 5,000 bp in old age.
35	**D**	**High telomerase and maintained telomeres** — High telomerase lets them divide indefinitely, making it a drug target.
36	**A**	**Blackburn, Greider and Szostak** — Blackburn, Greider and Szostak.

37	B	**5'-AUGCCU-3'** — mRNA is complementary to the template, with U instead of T.
38	C	**Pol III** — Pol III makes tRNA and 5S rRNA.
39	C	**GT … AG** — All introns begin GT (5') and end AG (3').
40	D	**How often transcription starts** — Mutations there cut transcription starts 10–20 fold.
41	A	**Silencers** — Repressors bind silencers and reduce transcription.
42	D	**Cis-acting element** — Cis = DNA sequence; trans = the protein that binds it.
43	A	**TFIID** — TFIID = TBP + TAFs, binding the TATA box.
44	A	**Act as a bridge between TFIID and RNA pol II** — TFIIB brings RNA polymerase II to the promoter-bound TFIID.
45	C	**AAUAAA** — Cutting occurs 10–35 nt downstream of AAUAAA.

Lecture 3

46	A	**7-methylguanosine** — A 7-methylguanosine cap is added early, when the transcript is 20–25 nt long.
47	B	**P site** — A = incoming aminoacyl-tRNA, P = peptidyl-tRNA, E = empty tRNA leaving.
48	D	**Third base** — Only the third base is flexible; the first two must pair exactly.
49	D	**Lowering histone basicity so DNA loosens** — Acetylation removes positive charge, freeing DNA and exposing the promoter.
50	C	**Histone deacetylase (HDAC)** — HDAC removes acetyl groups, restoring tight nucleosomes.
51	D	**hnRNA (pre-mRNA)** — Heterogeneous nuclear RNA is processed before leaving the nucleus.
52	C	**Small nuclear RNA + proteins** — They are the subunits of the spliceosome.
53	D	**Adding the poly-A tail** — Polyadenylation is a separate enzymatic step.
54	A	**One gene to make several proteins** — Different exon combinations give different mature mRNAs.
55	B	**20–25 nt long** — Capping happens early, during transcription by pol II.
56	B	**Protects mRNA from nucleases and helps ribosome binding** — It also aids nuclear export and splicing.
57	B	**Added enzymatically after transcription** — It isn't in the gene; it's added after cleavage downstream of AAUAAA.
58	B	**80 nt** — A single strand of about 80 nucleotides folded into an L shape.
59	A	**ATP** — ATP → AMP + PPi charges the tRNA.
60	D	**C or U** — Flexible pairing at the third codon base only.
61	C	**40% rRNA, 60% protein** — rRNA 40% and protein 60% per the lecture.
62	B	**The empty tRNA about to leave** — A = incoming, P = peptidyl, E = exit.
63	D	**N-terminus to C-terminus** — Methionine (N-terminus) first, last amino acid at the C-terminus.
64	D	**Codon recognition and translocation** — GTP powers aminoacyl-tRNA entry and ribosome movement.
65	B	**Hydrolysis of the bond to the P-site tRNA** — No tRNA reads stop codons; water releases the chain.
66	C	**Polysome** — Polyribosomes make many copies quickly.

Lecture 4

67	C	**Prolyl-hydroxylated, then recognized by VHL and ubiquitinated** — Prolyl hydroxylation lets VHL bind, ubiquitin is added, and the proteasome destroys HIF-1α (Nobel 2019).
68	C	**Ser, Thr and Tyr** — Kinases add phosphate to serine, threonine or tyrosine.
69	C	**61** — 61 code amino acids; 3 are stop codons.
70	D	**Alternative splicing plus PTMs** — Splicing expands the transcriptome; PTMs multiply the proteome.
71	C	**Polypeptides** — Ubiquitin is a small protein (polypeptide) added to the target.
72	C	**Complex groups** — Complex groups: glycosylation, AMPylation, lipidation.
73	C	**Negatively charged and hydrophilic** — Phosphate adds negative charge, changing conformation.
74	A	**Inhibited** — It is inhibited by phosphorylation and activated by dephosphorylation.
75	A	**Activated** — Several kinases phosphorylate and activate this tumour suppressor.
76	B	**Collagen** — Hydroxylases act on Pro and Lys in collagen.
77	C	**Three times** — Lys: 1–3 times; Arg: 1–2 times.
78	B	**8.6 kDa** — It is a small 8.6 kDa regulatory protein.

79	A	**One of seven lysines or the N-terminal Met of the previous ubiquitin** — That is how chains are built.
80	B	**Proteolysis** — Proteolysis = cleavage of proteins.
81	D	**Vitamin K** — γ-glutamyl carboxylase is vitamin K dependent.
82	C	**Accumulates and switches on genes like EPO** — Without prolyl hydroxylation, VHL can't tag it, so it builds up.
83	D	**VHL** — VHL binds it, leading to ubiquitination and proteasomal degradation.
84	B	**Met-Phe-Gly-Stop** — AUG Met, UUU Phe, GGC Gly, UAA stop.
85	A	**UGG** — UGG codes for tryptophan.
86	B	**Redundancy (degeneracy) of the code** — Several codons can specify the same amino acid.

Lecture 5

87	D	**3'UTR of the mRNA** — miRNA, inside RISC, pairs with the 3'UTR and causes degradation or blocks translation.
88	B	**Represses transcription** — It blocks Sp1 binding and recruits methyl-CpG binding protein + HDAC, closing chromatin.
89	A	**Constitutive expression** — Always on, at a constant level.
90	A	**Do not alter the DNA sequence** — Tools: cis/trans elements, ncRNAs, histone modifications.
91	C	**Six** — Six members that recruit co-activators.
92	A	**GC box** — Sp1 binds GC boxes and recruits TFIID and RNA pol II.
93	B	**DNA-binding and regulatory** — One attaches to DNA; the other interacts with other regulators.
94	A	**Transmit signals from activators to basal factors** — They link activators and the basal machinery.
95	C	**The DNA loops, bringing the enhancer-bound activator near the promoter** — Looping allows control at a distance.
96	B	**Represses transcription** — It blocks Sp1 and recruits methyl-CpG binding protein + HDAC.
97	D	**HDAC** — HDAC closes the chromatin.
98	C	**90%** — About 90% is removed to make mature mRNA.
99	B	**22 nt** — Single-stranded, about 22 nucleotides.
100	A	**Dicer** — Dicer cleaves double-stranded RNA.
101	B	**RISC** — RNA-induced silencing complex.
102	C	**RNA pol II** — Pol II makes the hairpin precursor.
103	B	**Decrease telomerase** — More miRNA silences more telomerase mRNA.
104	A	**microRNA** — Ambros and Ruvkun discovered microRNA.

Lecture 6

105	C	**Acetylated histone tails** — Bromodomains read acetylation; chromodomains read methylation.
106	A	**Acting as miRNA sponges** — circRNAs bind and sequester miRNAs, freeing those miRNAs' target mRNAs.
107	D	**CENP-A** — CENP-A nucleosomes are built into the kinetochore that attaches to the spindle.
108	B	**Back-splicing joining 5' to 3' end** — The cut segment closes into a ring.
109	B	**48 hours** — About 48 h, roughly 5× more stable than linear mRNA.
110	D	**They have no free ends for exonucleases** — Most RNA-degrading enzymes attack ends.
111	C	**Longer than 200 nt, often 1,000–10,000 nt** — They resemble mRNA: pol II, capped, polyadenylated.
112	B	**Decoy** — Decoys sequester proteins, silencing genes.
113	C	**Recruits HDAC to the ubiquitin promoter** — Deacetylation closes chromatin and lowers transcription.
114	B	**Decrease** — More free miR-236 silences p21 mRNA.
115	B	**H1** — H1 binds outside the octamer.
116	C	**Those near the ends** — DNA near the ends unwraps most often.
117	B	**ATP** — They slide, transfer or remodel nucleosomes using ATP.
118	C	**Transfer** — Sliding moves along the same DNA; transfer moves to another.
119	A	**Faces the octamer** — A:T-rich DNA bends easily toward the minor groove.
120	B	**Active chromatin** — Like acetylation, it reduces positive charge.

121	**B**	**Methylated tails** — Bromo = acetyl; chromo = methyl.
122	**A**	**DNA replication fidelity** — The four levels: transcription, RNA processing, translation, post-translation.
123	**A**	**Embryonic** — Embryo highest, fetus high, adult moderate.
124	**B**	**2α + 2β** — Fetal is 2α + 2γ; embryonic 2ζ + 2ε.
125	**B**	**10–15 nt** — Short sites of about 10–15 nucleotides.
126	**C**	**Individual base pairs can be distinguished there** — The major groove is wider and exposes sequence information.
127	**C**	**Two subunits joined by hydrophobic leucines into a Y shape** — The two helical arms fit into the major groove.

Lecture 7

128	**D**	**2022** — The T2T consortium filled the last gaps in March 2022.
129	**C**	**Zebrafish** — The list was E. coli, yeast, fruit fly and mouse.
130	**B**	**6** — 20 centres in 6 countries.
131	**A**	**Sanger sequencing** — Sanger sequencing, greatly advanced by innovations.
132	**D**	**More than 150,000** — Over 150,000 gaps; the 2003 version had fewer than 400.
133	**C**	**92%** — About 92% with fewer than 400 gaps.
134	**C**	**Telomeres and centromeres** — Highly repetitive DNA is hard to assemble.
135	**D**	**31 March 2022** — By the Telomere-to-Telomere (T2T) consortium.
136	**A**	**March 2021** — March 2021.
137	**D**	**Studying genomes of ancient Egyptians** — Also a reference Egyptian genome, common disease genes and sports performance.
138	**C**	**99.9%** — The 0.1% difference shapes traits and disease risk.
139	**B**	**SNP** — ≥ 1% = SNP; < 1% = mutation.
140	**C**	**At nucleotide 45, normal C is replaced by T** — Standard notation for a single-base change.